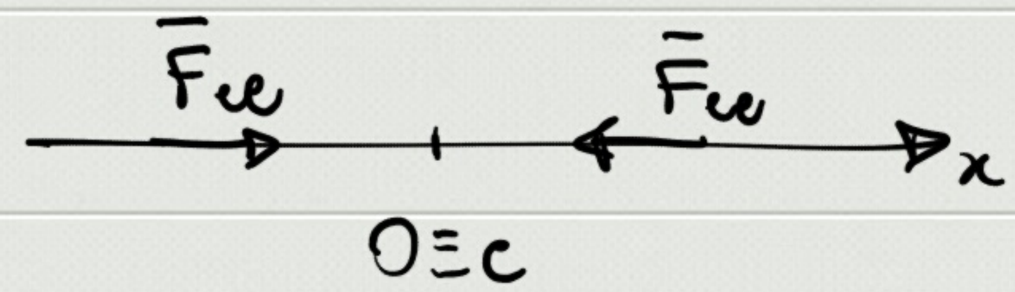
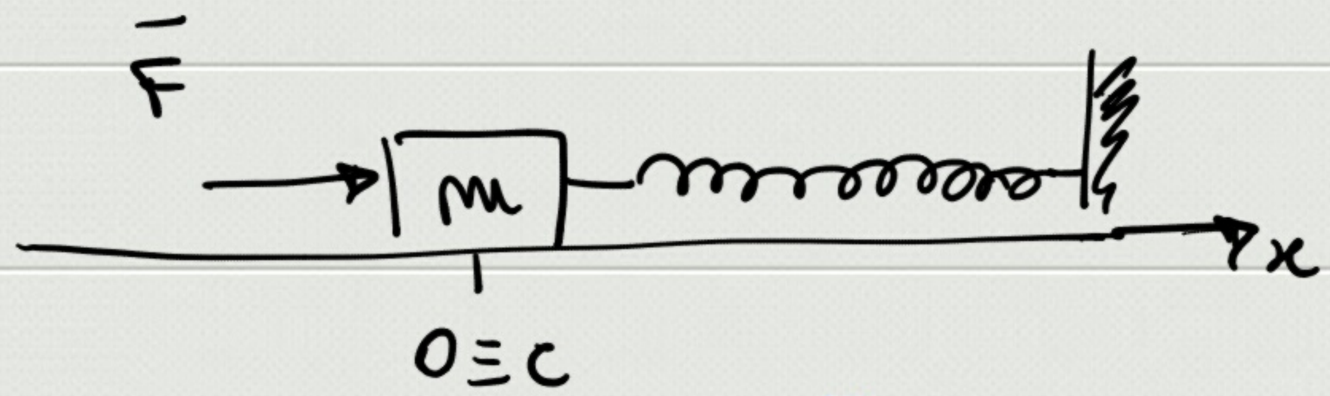
$$\bar{F}_{el} = -kx\,\bar{u}_x$$



$$W_{A\to B} = \int_A^B \bar{F}_{el}\, d\bar{s} = \int_{x_A}^{x_B} -kx\, dx = -\frac{1}{2}kx^2\Big|_{x_A}^{x_B} =$$

$$= -\left(\frac{1}{2}k x_B^2 - \frac{1}{2}k x_A^2\right) = -\Delta E_{p,el}$$

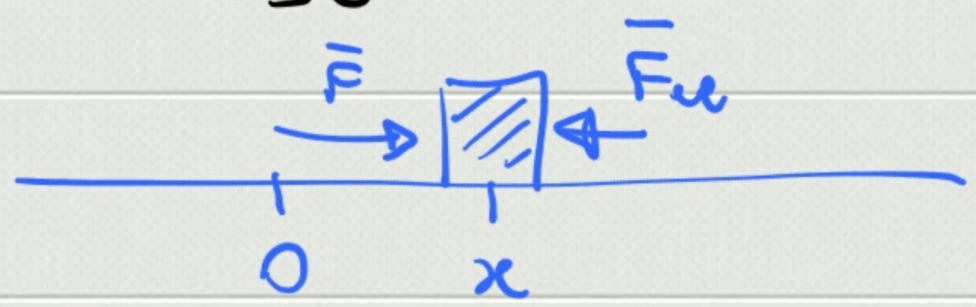
$$\boxed{E_{p,el} = \frac{1}{2}kx^2}$$



$\bar{F} = F\,\bar{u}_x \quad (F>0)$

$v = v(x)$

$x^*: \; v(x^*) = 0$



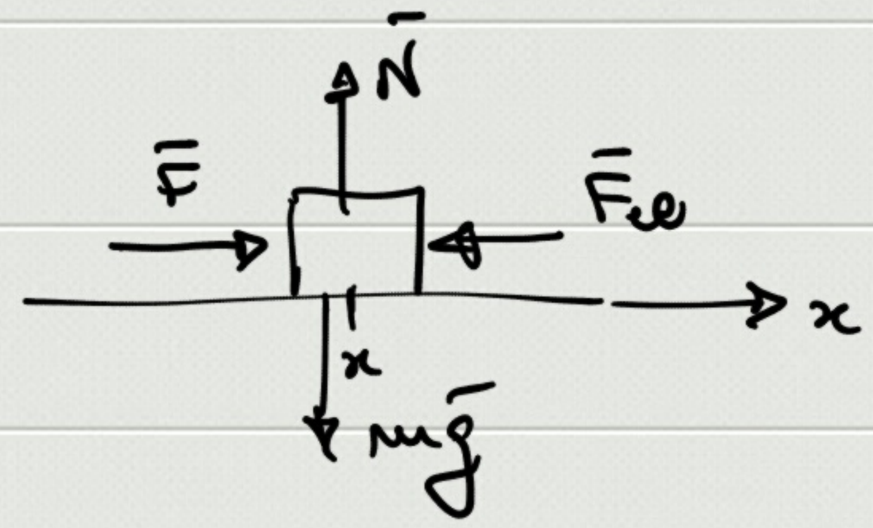
$$F - kx = ma = m\frac{d^2x}{dt^2}$$

$$\Rightarrow \quad \frac{d^2x}{dt^2} + \frac{k}{m}x = \frac{F}{m}$$

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

$$\Rightarrow F - kx = ma = mv\frac{dv}{dx} \Rightarrow \int (F - kx)\,dx = \int mv\,dv$$

$$W_{TOT,\,0\to x} = ?$$



$$= W_{F,\,0\to x} + W_{F_{el},\,0\to x} =$$

$$E_{p,F} = -Fx$$

$$= -\Delta E_{p,F} - \Delta E_{p,el} =$$

$$E_{p,el} = \frac{1}{2}kx^2$$

$$= -\left(-Fx - \emptyset\right) - \left(\frac{1}{2}kx^2 - \emptyset\right) = Fx - \frac{1}{2}kx^2$$

$$W_{TOT} = \Delta E_k = \frac{1}{2}mv^2(x) - \cancel{\frac{1}{2}mv_0^2}$$

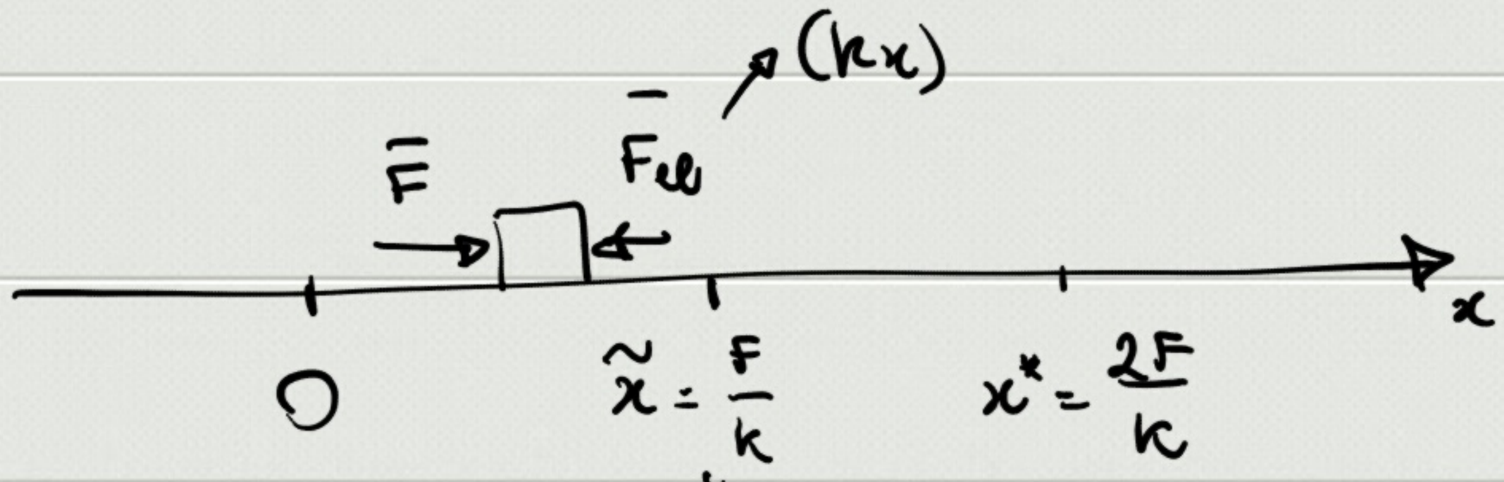
$$0$$

$$\frac{1}{2}mv^2(x) = Fx - \frac{1}{2}kx^2$$

$$\boxed{v(x) = \sqrt{\frac{x}{m}\left(2F - kx\right)}}$$

$$v(x^*) = 0 \quad \Longrightarrow \quad x^* = 0$$

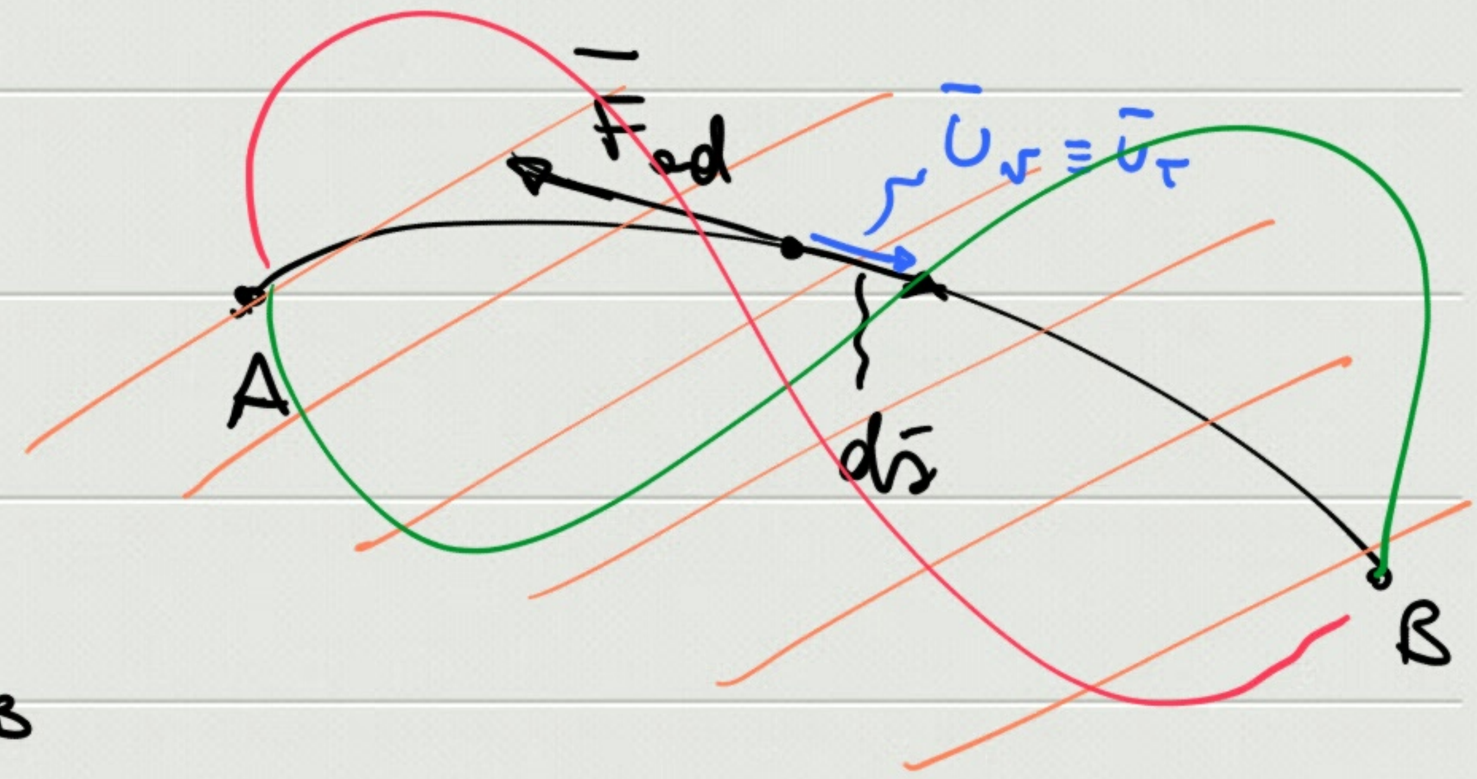
$$\boxed{x^* = \frac{2F}{k}}$$



$$\Downarrow$$

$$F = k\tilde{x}$$

$$|F_{el}|$$



$$W_{A\to B} = \int_A^B \bar{F}_{ad}\, d\bar{s} = \int_A^B \left(-\mu_d N \bar{u}_v\right) d\bar{s} = \qquad \bar{u}_v = \bar{u}_\tau$$

$$= -\mu_d N \int_A^B ds$$